

Proposition 1.7.3.

- (i) Every TX is also an IX (Fig. 1.7.4); thus, every topological minor of a graph is also its (ordinary) minor.
- (ii) If $\Delta(X) \leq 3$, then every IX contains a TX ; thus, every minor with maximum degree at most 3 of a graph is also its topological minor. $\square$

[4.4.2]
[7.3.1]
[12.7.3]

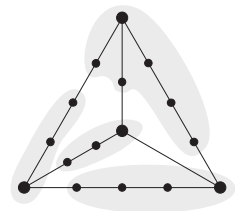


Fig. 1.7.4. A subdivision of K^4 viewed as an IK^4

Now that we have met all the standard relations between graphs, we can also define what it means to embed one graph in another. Basically, an *embedding* of G in H is an injective map $\varphi: V(G) \rightarrow V(H)$ that preserves the kind of structure we are interested in. Thus, φ embeds G in H ‘as a subgraph’ if it preserves the adjacency of vertices, and ‘as an induced subgraph’ if it preserves both adjacency and non-adjacency. If φ is defined on $E(G)$ as well as on $V(G)$ and maps the edges xy of G to independent paths in H between $\varphi(x)$ and $\varphi(y)$, it embeds G in H ‘as a topological minor’. Similarly, an embedding φ of G in H ‘as a minor’ would be a map from $V(G)$ to disjoint connected vertex sets in H (rather than to single vertices) so that H has an edge between the sets $\varphi(x)$ and $\varphi(y)$ whenever xy is an edge of G . Further variants are possible; depending on the context, one may wish to define embeddings ‘as a spanning subgraph’, ‘as an induced minor’ and so on, in the obvious way.

embedding

1.8 Euler tours

Any mathematician who happens to find himself in the East Prussian city of Königsberg (and in the 18th century) will lose no time to follow the great Leonhard Euler’s example and inquire about a round trip through the old city that traverses each of the bridges shown in Figure 1.8.1 exactly once.

Thus inspired,⁹ let us call a closed walk in a graph an *Euler tour* if it traverses every edge of the graph exactly once. A graph is *Eulerian* if it admits an Euler tour.

Eulerian

⁹ Anyone to whom such inspiration seems far-fetched, even after contemplating Figure 1.8.2, may seek consolation in the *multigraph* of Figure 1.10.1.

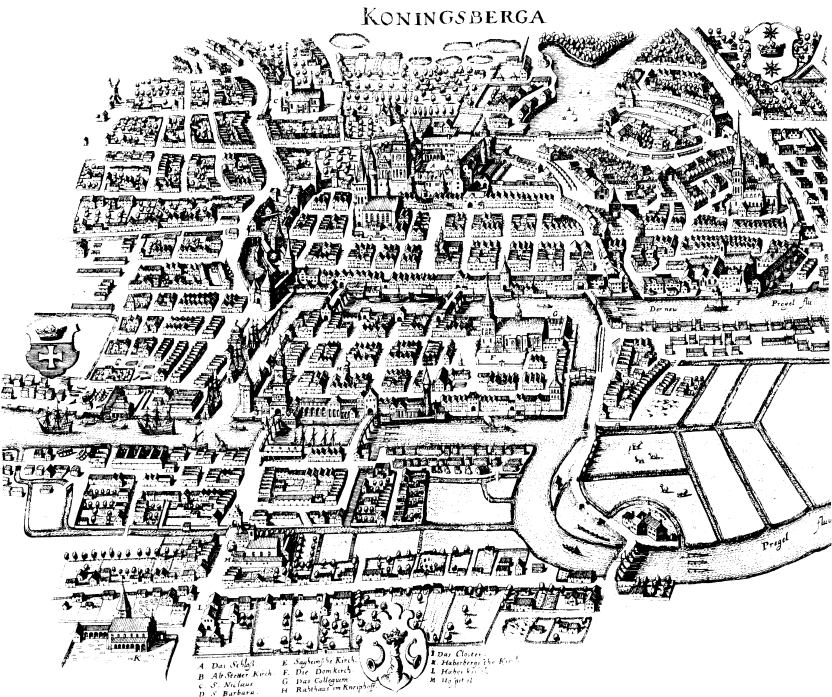


Fig. 1.8.1. The bridges of Königsberg (anno 1736)

[2.1.5]
[10.3.1]

Theorem 1.8.1. (Euler 1736)
A connected graph is Eulerian if and only if every vertex has even degree.

Proof. The degree condition is clearly necessary: a vertex appearing k times in an Euler tour (or $k + 1$ times, if it is the starting and finishing vertex and as such counted twice) must have degree $2k$.

Conversely, we show by induction on $\|G\|$ that every connected graph G with all degrees even has an Euler tour. The induction starts trivially with $\|G\| = 0$. Now let $\|G\| \geq 1$. Since all degrees are even, we can find in G a non-trivial closed walk that contains no edge more than once. (How exactly?) Let W be such a walk of maximal length,

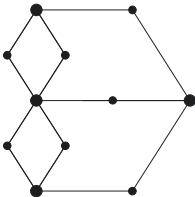


Fig. 1.8.2. A graph formalizing the bridge problem

and write F for the set of its edges. If $F = E(G)$, then W is an Euler tour. Suppose, therefore, that $G' := G - F$ has an edge.

For every vertex $v \in G$, an even number of the edges of G at v lies in F , so the degrees of G' are again all even. Since G is connected, G' has an edge e incident with a vertex on W . By the induction hypothesis, the component C of G' containing e has an Euler tour. Concatenating this with W (suitably re-indexed), we obtain a closed walk in G that contradicts the maximal length of W . $\square$

1.9 Some linear algebra

[8.7]

Let $G = (V, E)$ be a graph with n vertices and m edges, say $V = \{v_1, \dots, v_n\}$ and $E = \{e_1, \dots, e_m\}$. The *vertex space* $\mathcal{V}(G)$ of G is the vector space over the 2-element field $\mathbb{F}_2 = \{0, 1\}$ of all functions $V \rightarrow \mathbb{F}_2$. Every element of $\mathcal{V}(G)$ corresponds naturally to a subset of V , the set of those vertices to which it assigns a 1, and every subset of V is uniquely represented in $\mathcal{V}(G)$ by its indicator function. We may thus think of $\mathcal{V}(G)$ as the power set of V made into a vector space: the sum $U + U'$ of two vertex sets $U, U' \subseteq V$ is their symmetric difference (why?), and $U = -U$ for all $U \subseteq V$. The zero in $\mathcal{V}(G)$, viewed in this way, is the empty (vertex) set $\emptyset$. Since $\{\{v_1\}, \dots, \{v_n\}\}$ is a basis of $\mathcal{V}(G)$, its *standard basis*, we have $\dim \mathcal{V}(G) = n$.

 $G = (V, E)$ vertex
space $\mathcal{V}(G)$

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In the same way as above, the functions $E \rightarrow \mathbb{F}_2$ form the *edge space* $\mathcal{E}(G)$ of G : its elements correspond to the subsets of E , vector addition amounts to symmetric difference, $\emptyset \subseteq E$ is the zero, and $F = -F$ for all $F \subseteq E$. As before, $\{\{e_1\}, \dots, \{e_m\}\}$ is the *standard basis* of $\mathcal{E}(G)$, and $\dim \mathcal{E}(G) = m$. Given two elements F, F' of the edge space, viewed as functions $E \rightarrow \mathbb{F}_2$, we write

edge space
 $\mathcal{E}(G)$ standard
basis

$$\langle F, F' \rangle := \sum_{e \in E} F(e)F'(e) \in \mathbb{F}_2.$$

 $\langle F, F' \rangle$

This is zero if and only if F and F' have an even number of edges in common; in particular, we can have $\langle F, F \rangle = 0$ with $F \neq \emptyset$. Given a subspace $\mathcal{F}$ of $\mathcal{E}(G)$, we write

$$\mathcal{F}^\perp := \{ D \in \mathcal{E}(G) \mid \langle F, D \rangle = 0 \text{ for all } F \in \mathcal{F} \}.$$

 $\mathcal{F}^\perp$

This is again a subspace of $\mathcal{E}(G)$ (the space of all vectors solving a certain set of linear equations – which?), and one can show that

$$\dim \mathcal{F} + \dim \mathcal{F}^\perp = m.$$